Bluehost VPS Deployment & Release Guide

EASYPANEL DOCKER MULTI-STAGE PRODUCTION **Server IP:** 129.121.128.205 • **Project:** jit_support

Project: Jishnu Infotech Support System (`jit_support`)

Production Server: Bluehost Linux VPS (`129.121.128.205`)

Management Dashboard: Easypanel (`http://129.121.128.205:3000`)

Repository: `https://github.com/Jishnu-Support/jishnu-infotech-support.git` (Branch: `main`)

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1. Architecture & Deployment Workflow

```
flowchart LR
    LocalDev[Local Windows / Git] -->|git push origin main| GitHub[GitHub Repo: Jishnu-Support]
    GitHub -->|Webhook / Re-deploy| Easypanel[Easypanel at 129.121.128.205:3000]
    subgraph Bluehost VPS Container Network
        Easypanel --> Traefik[Traefik / Nginx Reverse Proxy: SSL 443]
        Traefik --> AppContainer["App Container (github)
Node.js 24 + Express + Vite (Port 3001)"]
        AppContainer --> AttachmentVol[("Volume Mount
/app/data/attachments")]
        AppContainer --> PostgresContainer[("Postgres Container (jit)
Port 5432 (Private)")]
    end
```

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2. Routine Deployment: Updating the App (Ongoing Updates)

Whenever you make code changes locally on your Windows PC and want them live on the Bluehost server:

Step 1: Commit and Push to GitHub

In Windows PowerShell (inside `C:\Users\HP\.gemini\antigravity\scratch\jishnu-infotech-support`):

```
git status
git add .
git commit -m "Update feature / bug fix description"
git push origin main
```

Step 2: Trigger Deployment in Easypanel

1. Open your browser and navigate to Easypanel:

👉 **`http://129.121.128.205:3000`**

2. In the left menu, click **Projects** → select **`jit_support`**.

3. Click on the App service named **`github`**.

4. In the top-right corner, click the **Deploy** button (or **Rebuild**).

Step 3: Monitor Build & Deployment Logs

1. Click the **Deployments** tab inside the `github` service.

2. Click on the latest deployment run to view the build logs.

3. You will see Docker execute the multi-stage build:

- `npm ci` (Installs dependencies)

- `npm run build` (Compiles TypeScript & Vite assets to `/dist`)

- Container startup (`npm start` → `node server/index.js`)

4. Verify the deployment finishes with:

```text
```

Application listening on port 3001

Database schema verified / migrations applied

Deployment successful

```

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#### Step 4: Health Check Verification

Open your browser and visit:

```
http://129.121.128.205:3001/api/health
```

*(Or your custom domain e.g. `https://support.yourdomain.com/api/health`)*

Expected JSON response:

```
{"status":"ok"}
```

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3. Fresh Server Setup: Initial Deployment from Scratch

If you ever need to set up a brand new Bluehost VPS:

Step 1: Provision the VPS

- **OS / Plan:** Ubuntu / Debian LTS (Minimum 2 vCPU, 4GB RAM, 50GB NVMe storage).
- **Firewall Ports:**
- Allow Inbound TCP `80` (HTTP) and `443` (HTTPS).
- Allow Inbound TCP `3000` (Easypanel Admin).
- Restrict Inbound TCP `22` (SSH) to your trusted IP.
- **DO NOT** expose TCP `5432` (PostgreSQL must stay private).

Step 2: Install Easypanel (If Not Pre-installed)

Connect via SSH and install Docker + Easypanel:

```
curl -sSL https://get.easypanel.io | sh
```

Open `http://:3000` and create the initial admin account.

Step 3: Create the PostgreSQL Database Service

1. In Easypanel → **Projects** → **Create Project** → name it `jit_support`.
2. Click **New Service** → select **PostgreSQL**.
3. Name the service: `jit`.
4. Leave it internal (Do not enable public port exposure).
5. Once healthy, navigate to the **Credentials** page and copy the internal `DATABASE_URL` (format: `postgres://user:password@jit:5432/jit_support`).

Step 4: Create the Application Service

1. In the `jit_support` project → **New Service** → select **App**.
2. Service name: `github`.
3. Set **Source** to **Git Repository**:
- Repository: `https://github.com/Jishnu-Support/jishnu-infotech-support.git`
- Branch: `main`
- Build Method: **Dockerfile**
- Dockerfile path: `/Dockerfile`
```

Step 5: Configure Persistent Storage Volume

1. Inside the `github` App service → open the **Storage** tab.
2. Click **Add Volume**:
- **Mount Path:** `/app/data/attachments`
- **Volume Name:** `attachments`

(This ensures ticket attachments and uploaded user manuals are never lost when the container rebuilds).

Step 6: Set Production Environment Variables

In the `github` App service → open the **Environment** tab and enter:

```
NODE_ENV=production
PORT=3001
DATABASE_URL=postgres://jit:PASSWORD_FROM_CREDENTIALS@jit:5432/jit_support
ATTACHMENTS_PATH=/app/data/attachments
JWT_SECRET=GENERATE_64_CHAR_HEX_STRING_HERE
APP_BASE_URL=https://support.yourdomain.com

SEED_CONSULTANT_USERNAME=superadmin
SEED_CONSULTANT_PASSWORD=SuperAdminPassword123!
SEED_CONSULTANT_NAME=Jishnu Infotech Super Admin

SMTP_HOST=smtp.yourmailserver.com
SMTP_PORT=587
SMTP_USER=support@yourdomain.com
SMTP_PASS=your_email_password
SMTP_FROM="Jishnu Support "
```

Step 7: Configure Custom Domain & Automatic SSL

1. Inside the `github` App service → open the **Domains** tab.
2. Click **Add Domain**:
- Host: `support.yourdomain.com`
- Port: `3001`
- Enable HTTPS: **Checked** (Traefik automatically issues free Let's Encrypt SSL).
3. In your DNS provider (Cloudflare / Bluehost / GoDaddy), add an **A Record**:
- **Name:** `support`
- **Target / Value:** `129.121.128.205`

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4. Rollback & Emergency Recovery

If a deployment introduces a bug:

1. Open Easypanel → **Projects** → `jit_support` → `github`.
2. Click the **Deployments** tab.
3. Find the previous stable build and click **Rollback**.
4. If database corruption occurred, use the backup file created earlier:

```bash
gunzip -c /tmp/db_backup.sql.gz | psql -U $POSTGRES_USER -d $POSTGRES_DB
```

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5. Deployment Troubleshooting Matrix

Issue	Cause	Fix
Build fails at `npm run build`	TypeScript error or missing asset	Run `npm run build` locally on Windows first to catch errors before pushing.
App crashes on startup with `DATABASE_URL` error	Invalid DB connection string or Postgres container starting up	Check that `jit` Postgres service is in status "Running" and `DATABASE_URL` has correct credentials.
Uploaded files disappear after deployment	Missing volume mount	Verify `/app/data/attachments` is mapped to a named Docker volume in the **Storage** tab.
502 Bad Gateway error on domain	Node.js process failed to bind port `3001`	Check App service logs in Easypanel; ensure `PORT=3001` is defined in Environment settings.
Database changes not reflected	Migrations pending	The server runs `db.js` migrations automatically on startup; check the startup log for migration errors.